



WIPRO CAPSTONE PROJECT REPORT

Name	Gadi Teja Sai Manikanta
Superset ID	5475580
Organization	Wipro
Project Type	Capstone project
Project Name	Automation Testing using Selenium Pytest Framework
	Automation Testing using Behave BDD Framework
Domain	Healthcare
Module	Lab Test - Vital Organs

Synopsis

The Apollo Pharmacy Lab Test Automation Framework is a Selenium-based end-to-end web automation testing project developed using Python, Behave BDD, and Allure Reporting. The framework automates the Apollo Pharmacy Vital Organs Lab Test booking workflow, including login, package selection, slot booking, cart review, and payment flow.

The project follows the Page Object Model (POM) design pattern and supports:

- Behavior Driven Development (BDD)
- Data Driven Testing
- Allure Reporting Dashboard
- Logging and Screenshots
- Excel-based test data handling

The framework simulates real-world healthcare application automation testing.

Objectives of the Project

The main objectives of this project are:

- Automate Apollo Pharmacy Vital Organs lab test booking process
- Validate positive and negative test scenarios
- Implement enterprise-level Selenium framework
- Integrate Behave BDD for readable feature files
- Generate professional Allure Reports
- Perform data-driven testing using Excel
- Improve maintainability using POM architecture

Technologies Used

Technology	Purpose
Python	Programming Language
Selenium WebDriver	Browser Automation
Behave	BDD Framework
Allure Reports	Reporting Dashboard
Pandas	Excel Data Handling
OpenPyXL	Excel File Reading
ChromeDriver	Browser Driver
Pytest	Supporting Testing Library

Project Architecture

```
Apollo247Framework/  
├── tests/  
│   ├── test_search.py  
│   ├── test_login.py  
│   └── test_booking.py  
├── pages/  
│   ├── home_page.py  
│   ├── vital_organ_page.py  
│   └── checkout_page.py  
├── utils/  
│   ├── driver_factory.py  
│   ├── excel_reader.py  
│   └── waits.py  
├── testdata/  
│   └── testdata.xlsx  
├── reports/  
├── screenshots/  
├── requirements.txt  
└── pytest.ini
```

Page Object Model (POM)

Page Object Model is a Selenium design pattern where each webpage is represented as a separate class.

Advantages

- Better code organization
- Easy maintenance
- Reduced duplication
- Improved readability

Pages Implemented

- HomePage
- LabTestsPage
- VitalOrgansPage
- PaymentPage

Behave BDD Framework

Behavior Driven Development allows writing tests in plain English using Gherkin syntax.

Example:

Feature: Apollo Lab Test Booking

Scenario: Complete Booking

Given user launches Apollo application

When user navigates to lab tests

Then booking should complete successfully

Benefits

- Easy readability
- Business-friendly
- Better collaboration

- Structured automation

Feature Files

Three feature files were implemented:

Positive Scenarios

- Valid Thyroid Booking
- Valid Mobile Login
- Successful Slot Selection

Negative Scenarios

- Invalid Mobile Number
- Invalid Card Details

End-to-End Scenario

- Complete Lab Test Booking Flow

Step Definitions

Step definitions connect feature file steps with Python automation code.

Example:

```
@when('user navigates to lab tests')
```

```
def step_impl(context):
```

```
context.home.click_lab_tests()
```

Responsibilities

- Execute actions
- Validate outputs
- Handle test flow

Selenium Automation Workflow

The automated workflow includes:

1. Open Apollo Pharmacy
2. Navigate to Lab Tests
3. Select Vital Organs
4. Open Thyroid Package
5. Add Package to Cart
6. Login with Mobile Number
7. OTP Verification
8. Patient Selection
9. Slot Selection
10. Review Cart
11. Proceed to Payment
12. Enter Card Details
13. Click Pay

Data Driven Testing

The framework uses Excel-based external test data.

Excel Columns

package_name	mobile_number	card_name	card_number	expiry_date	cvv
Thyroid	9381866215	Teja Sai	4.11111E+15	1230	123
Liver	9381866215	Rahul	5.55556E+15	1128	456
Kidney	938186625	Ram	4.59612E+11	1129	541
Heart	9381866215	Manikanta	8.56943E+11	1033	126

- Easy maintenance
- Multiple test execution
- Reusability

Logging

05/21/2026 11:04:48 AM : INFO : ===== Navigation Test Started
=====

05/21/2026 11:04:51 AM : INFO : Opening Home Page

05/21/2026 11:04:55 AM : INFO : Home Page Verified

05/21/2026 11:05:06 AM : INFO : Lab Tests Clicked

05/21/2026 11:05:13 AM : INFO : Lab Tests Page Verified

05/21/2026 11:05:13 AM : INFO : ===== Navigation Test Passed
=====

05/21/2026 11:05:16 AM : INFO : Browser Closed

05/21/2026 11:05:16 AM : INFO : ===== Navigation Test Ended
=====

05/21/2026 11:05:49 AM : INFO : ===== TEST STARTED =====

05/21/2026 11:05:51 AM : INFO : Opening Home Page
05/21/2026 11:05:53 AM : INFO : Home Page Verified
05/21/2026 11:06:03 AM : INFO : Lab Tests Clicked
05/21/2026 11:06:10 AM : INFO : Lab Tests Page Verified
05/21/2026 11:06:11 AM : INFO : Vital Organs Selected
05/21/2026 11:06:12 AM : INFO : Vital Organs Page Verified
05/21/2026 11:06:12 AM : INFO : Thyroid Card Opened
05/21/2026 11:06:14 AM : INFO : Add Button Clicked
05/21/2026 11:06:24 AM : INFO : Go To Cart Clicked
05/21/2026 11:07:07 AM : INFO : Mobile Number Entered
05/21/2026 11:07:49 AM : INFO : OTP Verified
05/21/2026 11:07:59 AM : INFO : Go To Cart Clicked Again
05/21/2026 11:08:07 AM : INFO : Patient Checkbox Selected
05/21/2026 11:08:19 AM : INFO : Select Slot Clicked
05/21/2026 11:08:40 AM : INFO : Review Cart Clicked
05/21/2026 11:08:52 AM : INFO : Proceed To Pay Clicked
05/21/2026 11:08:59 AM : INFO : Credit/Debit Card Section Clicked
05/21/2026 11:11:40 AM : INFO : Card Details Entered
05/21/2026 11:12:02 AM : INFO : Pay Button Clicked
05/21/2026 11:12:03 AM : INFO : Success Screenshot Saved:
screenshots/Thyroid_success.png
05/21/2026 11:12:03 AM : INFO : ===== TEST PASSED =====
05/21/2026 11:12:03 AM : INFO : Closing Browser
05/21/2026 11:12:05 AM : INFO : ===== TEST ENDED =====
05/21/2026 11:18:08 AM : INFO : ===== Navigation Test Started
=====

05/21/2026 11:18:11 AM : INFO : Opening Home Page
05/21/2026 11:18:15 AM : INFO : Home Page Verified
05/21/2026 11:18:26 AM : INFO : Lab Tests Clicked
05/21/2026 11:18:37 AM : INFO : Lab Tests Page Verified
05/21/2026 11:18:37 AM : INFO : ===== Navigation Test Passed
=====

05/21/2026 11:18:40 AM : INFO : Browser Closed
05/21/2026 11:18:40 AM : INFO : ===== Navigation Test Ended
=====

05/21/2026 11:37:12 AM : INFO : ===== TEST STARTED =====
05/21/2026 11:37:14 AM : INFO : Opening Home Page
05/21/2026 11:37:15 AM : INFO : Home Page Verified
05/21/2026 11:37:25 AM : INFO : Lab Tests Clicked
05/21/2026 11:37:27 AM : INFO : Lab Tests Page Verified
05/21/2026 11:37:29 AM : INFO : Vital Organs Selected
05/21/2026 11:37:29 AM : INFO : Vital Organs Page Verified
05/21/2026 11:37:29 AM : INFO : Thyroid Card Opened
05/21/2026 11:37:30 AM : INFO : Add Button Clicked
05/21/2026 11:37:40 AM : INFO : Go To Cart Clicked
05/21/2026 11:38:24 AM : INFO : Mobile Number Entered
05/21/2026 11:39:06 AM : INFO : OTP Verified
05/21/2026 11:39:16 AM : INFO : Go To Cart Clicked Again
05/21/2026 11:39:23 AM : INFO : Patient Checkbox Selected
05/21/2026 11:39:30 AM : INFO : Select Slot Clicked
05/21/2026 11:39:51 AM : INFO : Review Cart Clicked
05/21/2026 11:40:03 AM : INFO : Proceed To Pay Clicked
05/21/2026 11:40:10 AM : INFO : Credit/Debit Card Section Clicked
05/21/2026 11:42:41 AM : INFO : Card Details Entered
05/21/2026 11:42:53 AM : INFO : Pay Button Clicked
05/21/2026 11:42:53 AM : INFO : Success Screenshot Saved:
screenshots/Thyroid_success.png
05/21/2026 11:42:53 AM : INFO : ===== TEST PASSED =====
05/21/2026 11:42:53 AM : INFO : Closing Browser
05/21/2026 11:42:59 AM : INFO : ===== TEST ENDED =====
05/22/2026 09:20:06 AM : INFO : ===== Navigation Test Started
=====

05/22/2026 09:20:15 AM : INFO : Opening Home Page
05/22/2026 09:20:16 AM : INFO : Home Page Verified
05/22/2026 09:20:27 AM : INFO : Lab Tests Clicked
05/22/2026 09:20:31 AM : INFO : Lab Tests Page Verified
05/22/2026 09:20:31 AM : INFO : ===== Navigation Test Passed
=====

05/22/2026 09:20:33 AM : INFO : Browser Closed

05/22/2026 09:20:33 AM : INFO : ===== Navigation Test Ended

=====

05/22/2026 09:20:57 AM : INFO : ===== TEST STARTED =====

05/22/2026 09:21:00 AM : INFO : Opening Home Page

05/22/2026 09:21:01 AM : INFO : Home Page Verified

05/22/2026 09:21:20 AM : INFO : Lab Tests Clicked

05/22/2026 09:21:20 AM : INFO : Lab Tests Page Verified

05/22/2026 09:21:20 AM : INFO : Vital Organs Selected

05/22/2026 09:21:21 AM : INFO : Vital Organs Page Verified

05/22/2026 09:21:21 AM : INFO : Thyroid Card Opened

05/22/2026 09:21:22 AM : INFO : Add Button Clicked

05/22/2026 09:21:32 AM : INFO : Go To Cart Clicked

05/22/2026 09:22:16 AM : INFO : Mobile Number Entered

05/22/2026 09:22:58 AM : INFO : OTP Verified

05/22/2026 09:23:08 AM : INFO : Go To Cart Clicked Again

05/22/2026 09:23:18 AM : ERROR : Failure Screenshot Saved:
screenshots/Thyroid_failure.png

05/22/2026 09:23:18 AM : ERROR : Test Failed: Message:

Stacktrace:

chromedriver!GetHandleVerifier [0x7ff6899a7de5+14895]

chromedriver!GetHandleVerifier [0x7ff6899a7e50+14900]

chromedriver!(No symbol) [0x7ff68970d5ad]

chromedriver!(No symbol) [0x7ff689767822]

chromedriver!(No symbol) [0x7ff689767b2c]

chromedriver!(No symbol) [0x7ff6897b7d17]

chromedriver!(No symbol) [0x7ff6897b486f]

chromedriver!(No symbol) [0x7ff689759df8]

chromedriver!(No symbol) [0x7ff68975ace3]

chromedriver!GetHandleVerifier [0x7ff689cbcc49+3296f9]

chromedriver!GetHandleVerifier [0x7ff689cb7375+323e25]

chromedriver!GetHandleVerifier [0x7ff689cd82+348732]

chromedriver!GetHandleVerifier [0x7ff6899c6045+32af5]

chromedriver!GetHandleVerifier [0x7ff6899cecec+3b79c]

chromedriver!GetHandleVerifier [0x7ff6899b1bc4+1e674]

chromedriver!GetHandleVerifier [0x7ff6899b1d54+1e804]

chromedriver!GetHandleVerifier [0x7ff6899960e7+2b97]
KERNEL32!BaseThreadInitThunk [0x7ffa69c6e8d7+17]
ntdll!RtlUserThreadStart [0x7ffa6ab2c3fc+2c]

05/22/2026 09:23:18 AM : INFO : Closing Browser
05/22/2026 09:23:21 AM : INFO : ===== TEST ENDED =====
05/22/2026 09:23:38 AM : INFO : ===== TEST STARTED =====
05/22/2026 09:23:41 AM : INFO : Opening Home Page
05/22/2026 09:23:45 AM : INFO : Home Page Verified
05/22/2026 09:24:02 AM : INFO : Lab Tests Clicked
05/22/2026 09:24:02 AM : INFO : Lab Tests Page Verified
05/22/2026 09:24:03 AM : INFO : Vital Organs Selected
05/22/2026 09:24:03 AM : INFO : Vital Organs Page Verified
05/22/2026 09:24:03 AM : INFO : Thyroid Card Opened
05/22/2026 09:24:04 AM : INFO : Add Button Clicked
05/22/2026 09:24:14 AM : INFO : Go To Cart Clicked
05/22/2026 09:24:58 AM : INFO : Mobile Number Entered
05/22/2026 09:25:40 AM : INFO : OTP Verified
05/22/2026 09:25:50 AM : INFO : Go To Cart Clicked Again
05/22/2026 09:25:57 AM : INFO : Patient Checkbox Selected
05/22/2026 09:26:04 AM : INFO : Select Slot Clicked
05/22/2026 09:26:25 AM : INFO : Review Cart Clicked
05/22/2026 09:26:33 AM : INFO : Proceed To Pay Clicked
05/22/2026 09:26:39 AM : INFO : Credit/Debit Card Section Clicked
05/22/2026 09:29:11 AM : INFO : Card Details Entered
05/22/2026 09:29:18 AM : INFO : Pay Button Clicked
05/22/2026 09:29:18 AM : INFO : Success Screenshot Saved:
screenshots/Thyroid_success.png
05/22/2026 09:29:18 AM : INFO : ===== TEST PASSED =====
05/22/2026 09:29:18 AM : INFO : Closing Browser
05/22/2026 09:29:21 AM : INFO : ===== TEST ENDED =====
05/22/2026 12:47:57 PM : INFO : ===== TEST STARTED =====
05/22/2026 12:48:01 PM : INFO : Opening Home Page
05/22/2026 12:48:08 PM : INFO : Home Page Verified
05/22/2026 12:48:18 PM : INFO : Lab Tests Clicked

05/22/2026 12:48:30 PM : INFO : Lab Tests Page Verified
05/22/2026 12:48:30 PM : INFO : Vital Organs Selected
05/22/2026 12:48:36 PM : INFO : Vital Organs Page Verified
05/22/2026 12:48:36 PM : INFO : Thyroid Card Opened
05/22/2026 12:48:38 PM : INFO : Add Button Clicked
05/22/2026 12:48:48 PM : INFO : Go To Cart Clicked
05/22/2026 12:49:32 PM : INFO : Mobile Number Entered
05/22/2026 12:50:14 PM : INFO : OTP Verified
05/22/2026 12:50:24 PM : INFO : Go To Cart Clicked Again
05/22/2026 12:50:31 PM : INFO : Patient Checkbox Selected
05/22/2026 12:50:38 PM : INFO : Select Slot Clicked
05/22/2026 12:50:59 PM : INFO : Review Cart Clicked
05/22/2026 12:51:06 PM : INFO : Proceed To Pay Clicked
05/22/2026 12:51:19 PM : INFO : Credit/Debit Card Section Clicked
05/22/2026 12:53:49 PM : INFO : Card Details Entered
05/22/2026 12:53:54 PM : INFO : Pay Button Clicked
05/22/2026 12:53:54 PM : INFO : Success Screenshot Saved:
screenshots/Thyroid_success.png
05/22/2026 12:53:54 PM : INFO : ===== TEST PASSED =====
05/22/2026 12:53:54 PM : INFO : Closing Browser
05/22/2026 12:53:56 PM : INFO : ===== TEST ENDED =====

Screenshots

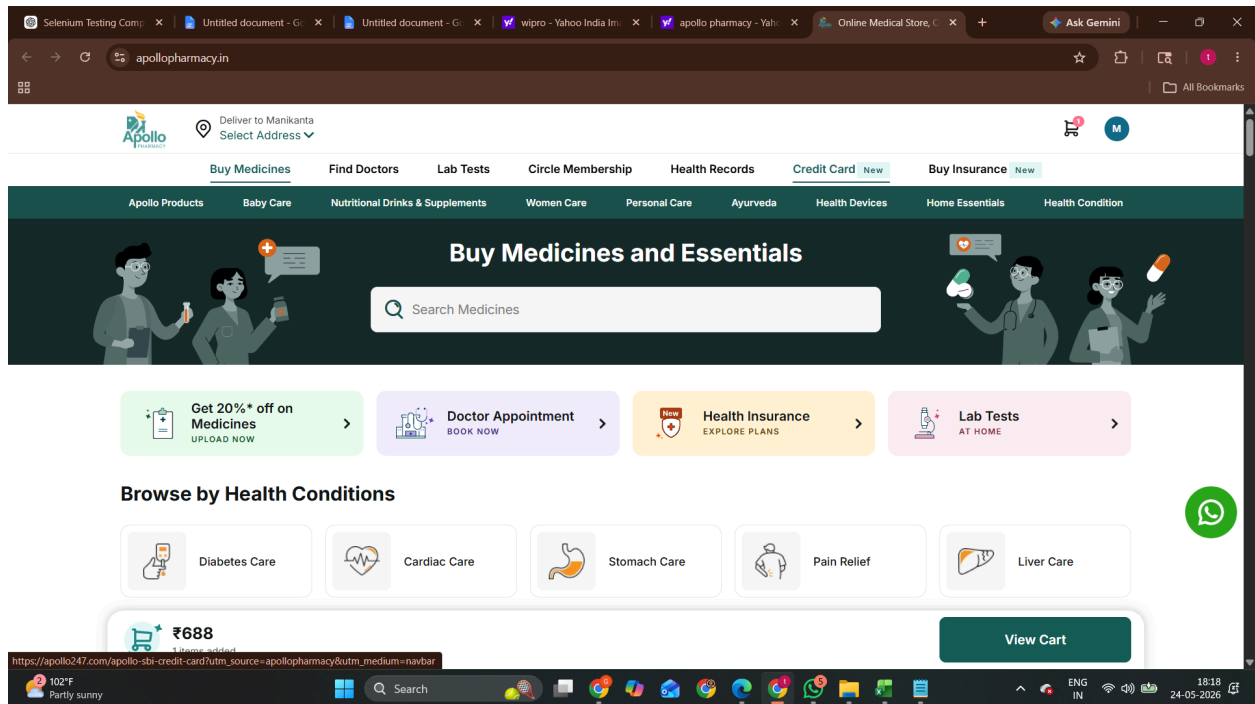


Fig 1: home page

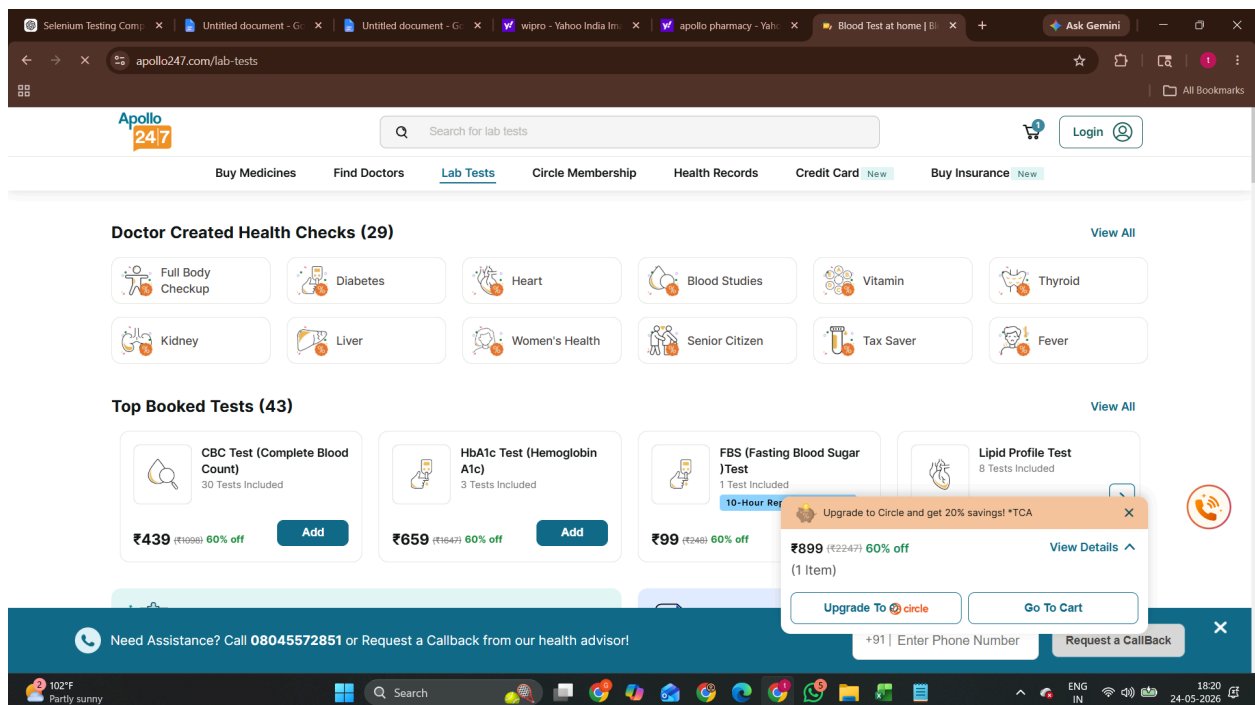


Fig2: Lab Test page

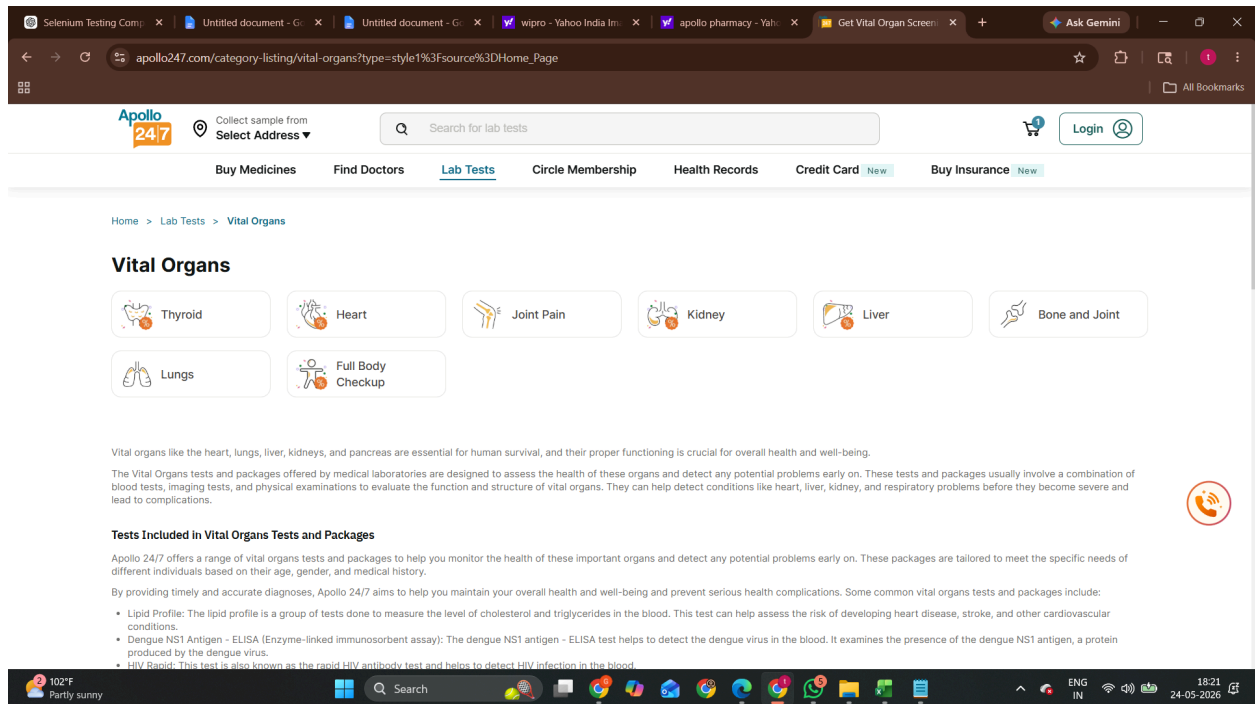


Fig3: Vital Organs page

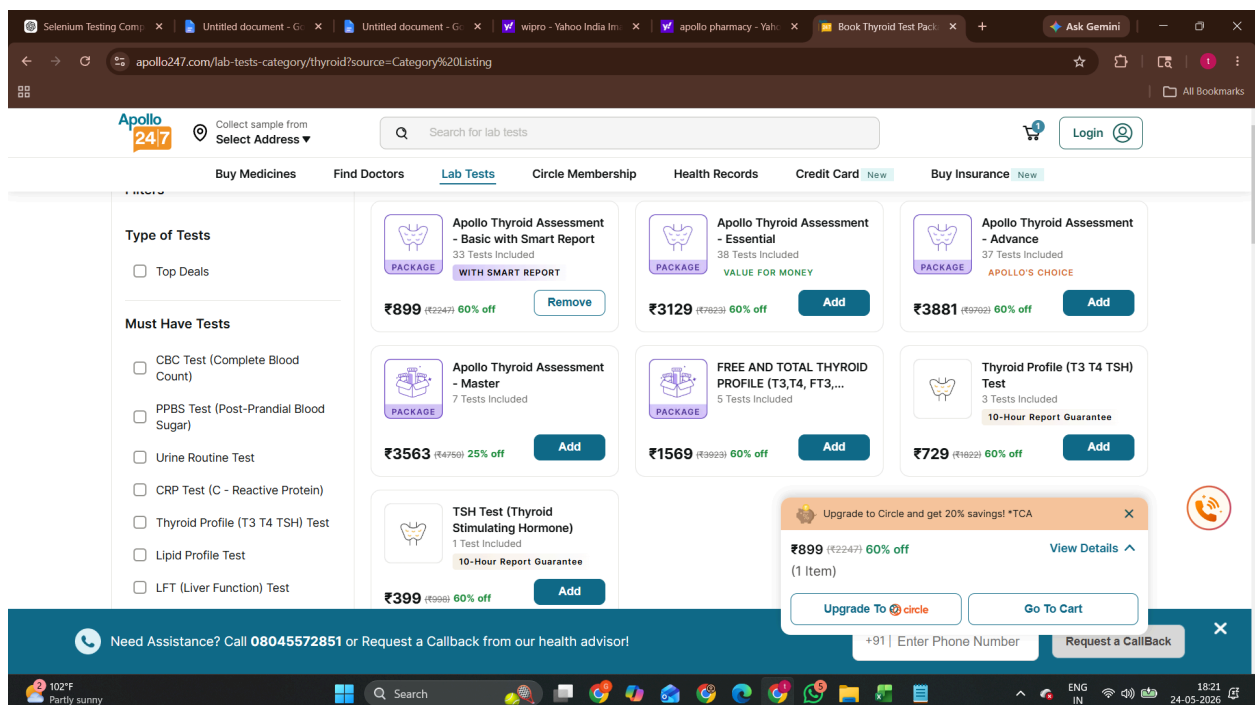


Fig4: Package Added

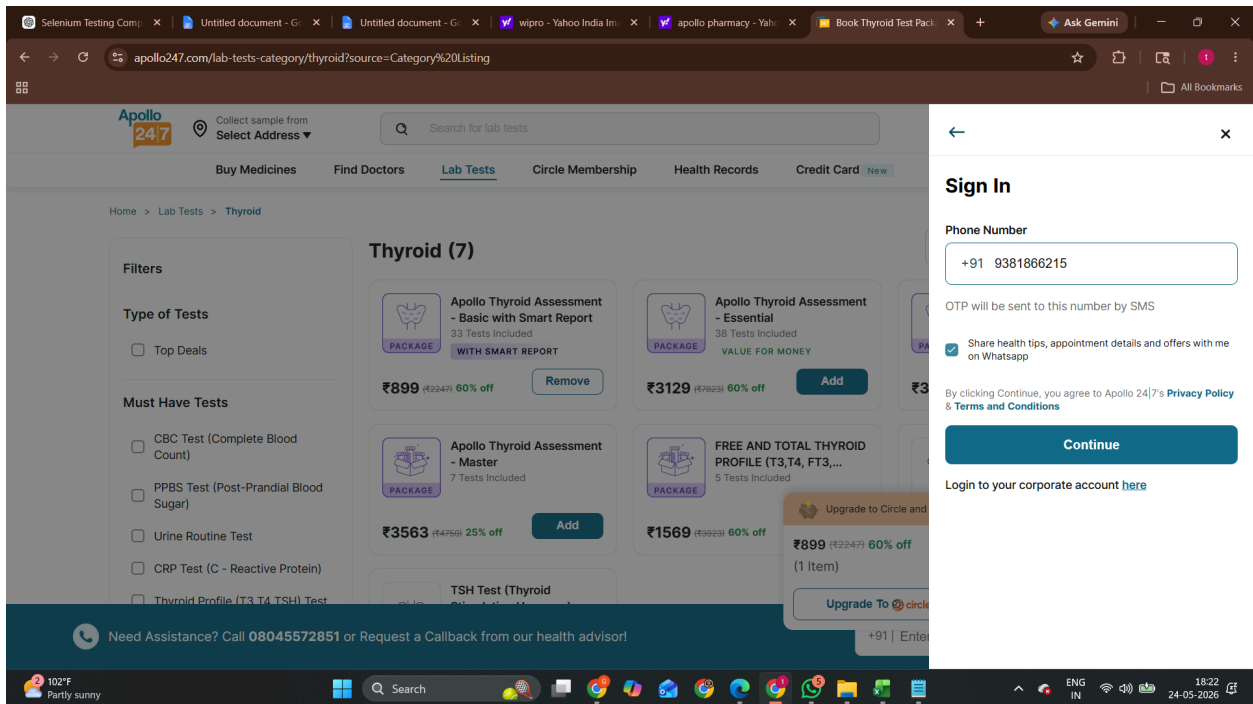


Fig5: Login

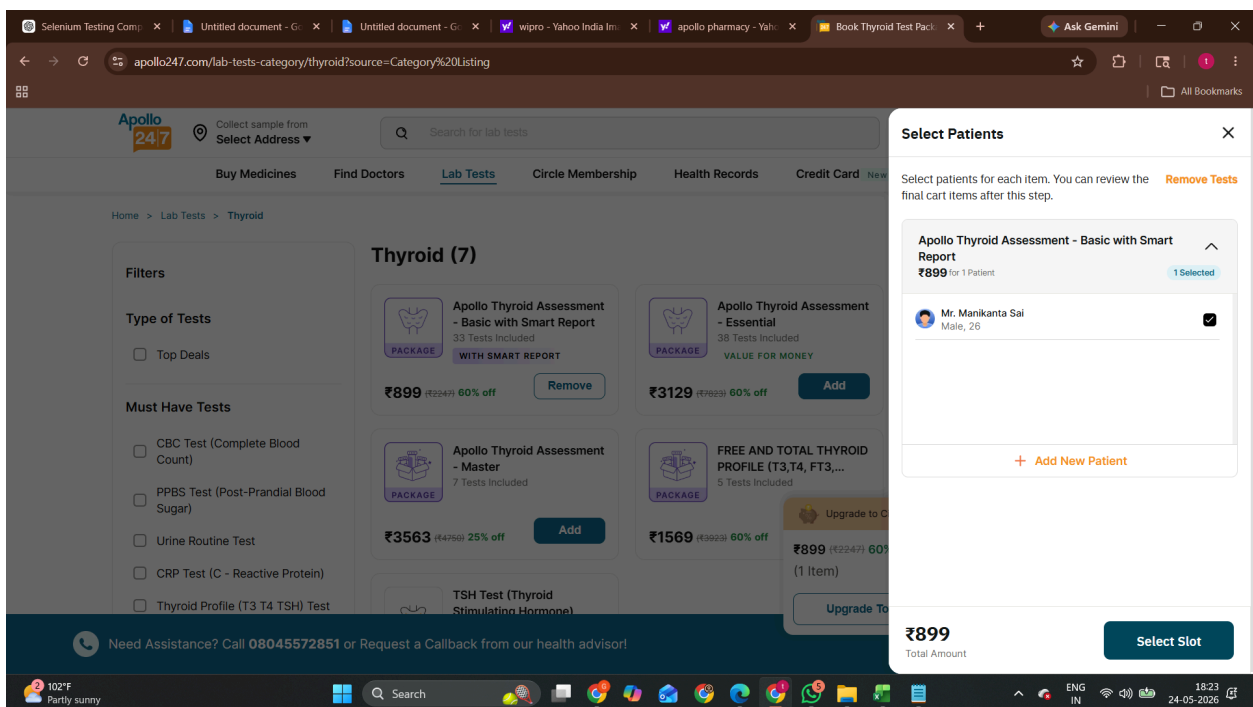


Fig6: Selecting Patient

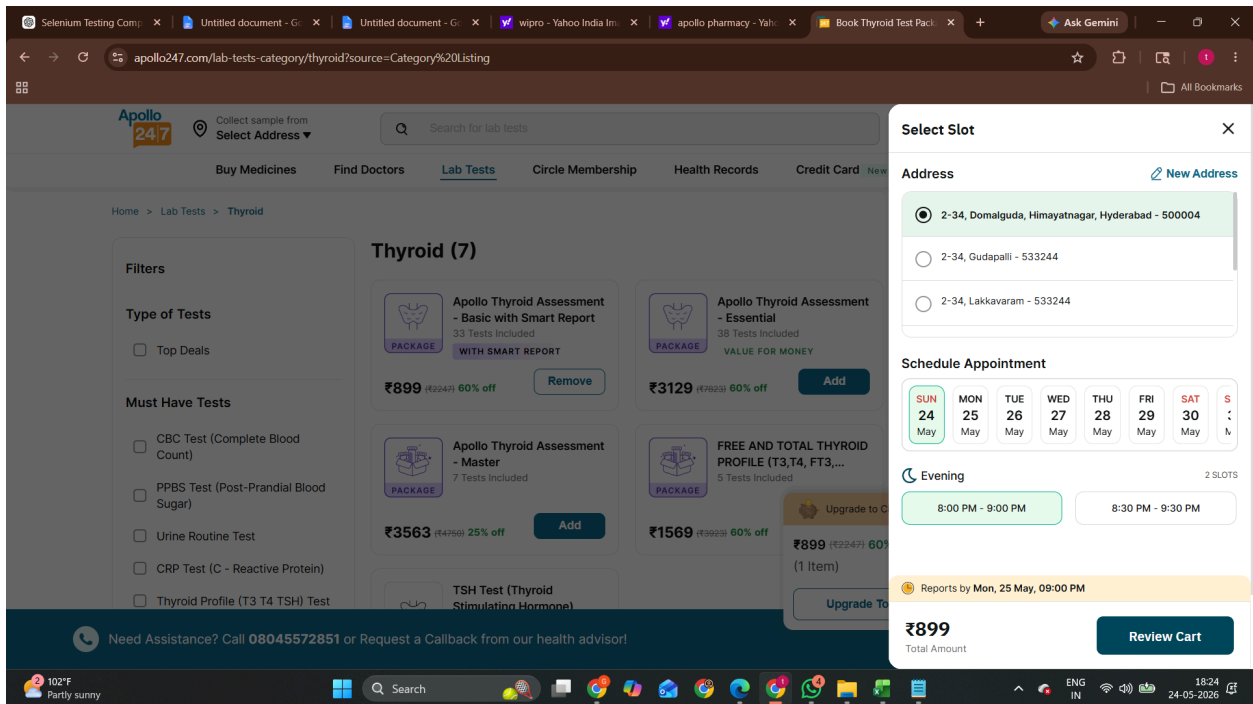


Fig7: Slot selection

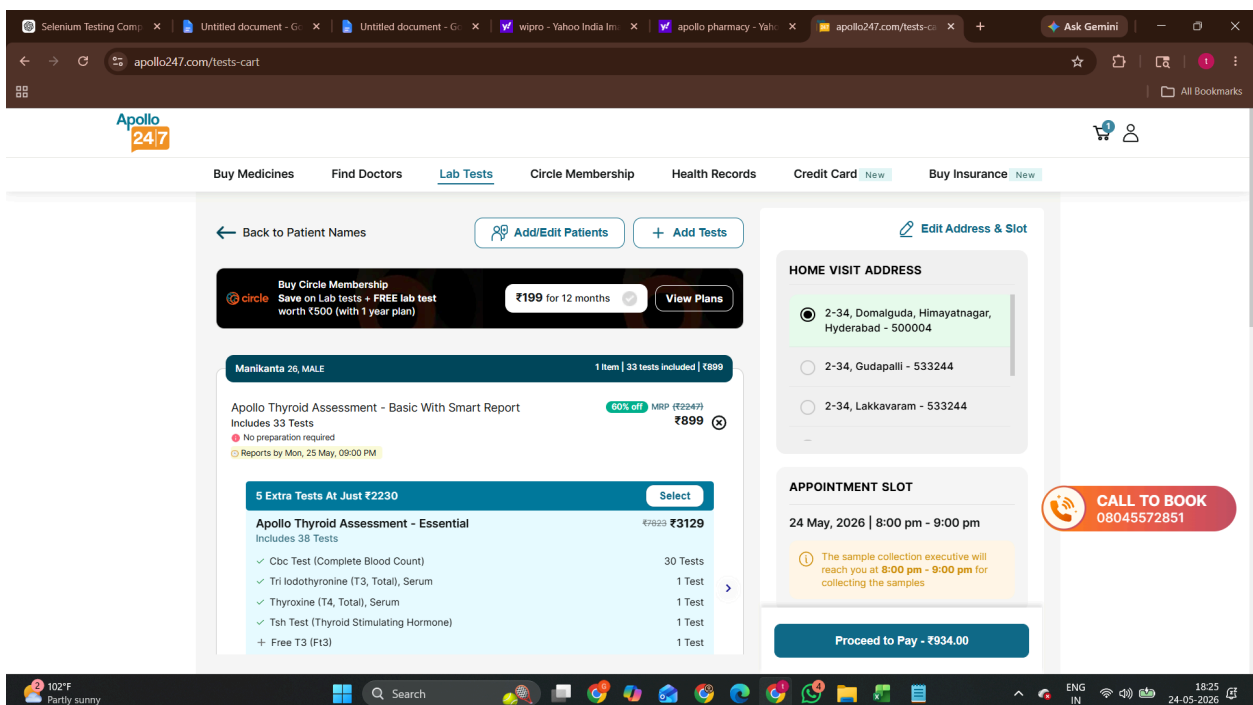


Fig8: Proceed to pay

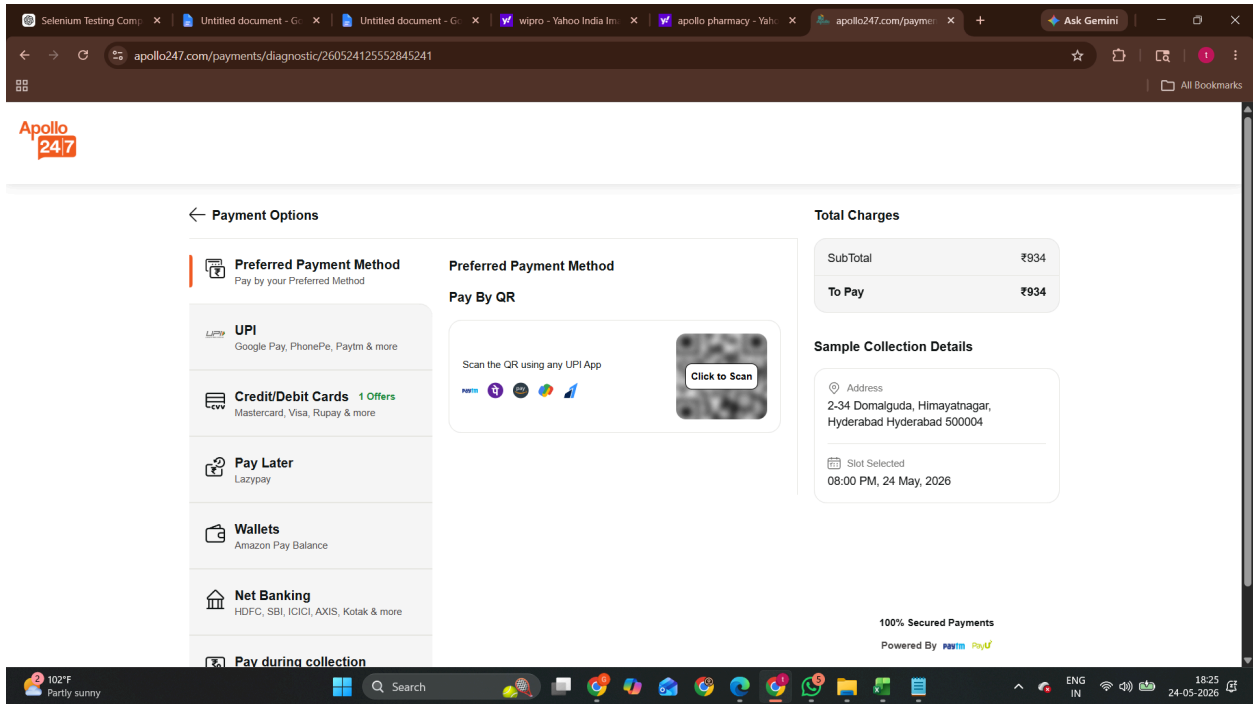


Fig9: Payment gateway

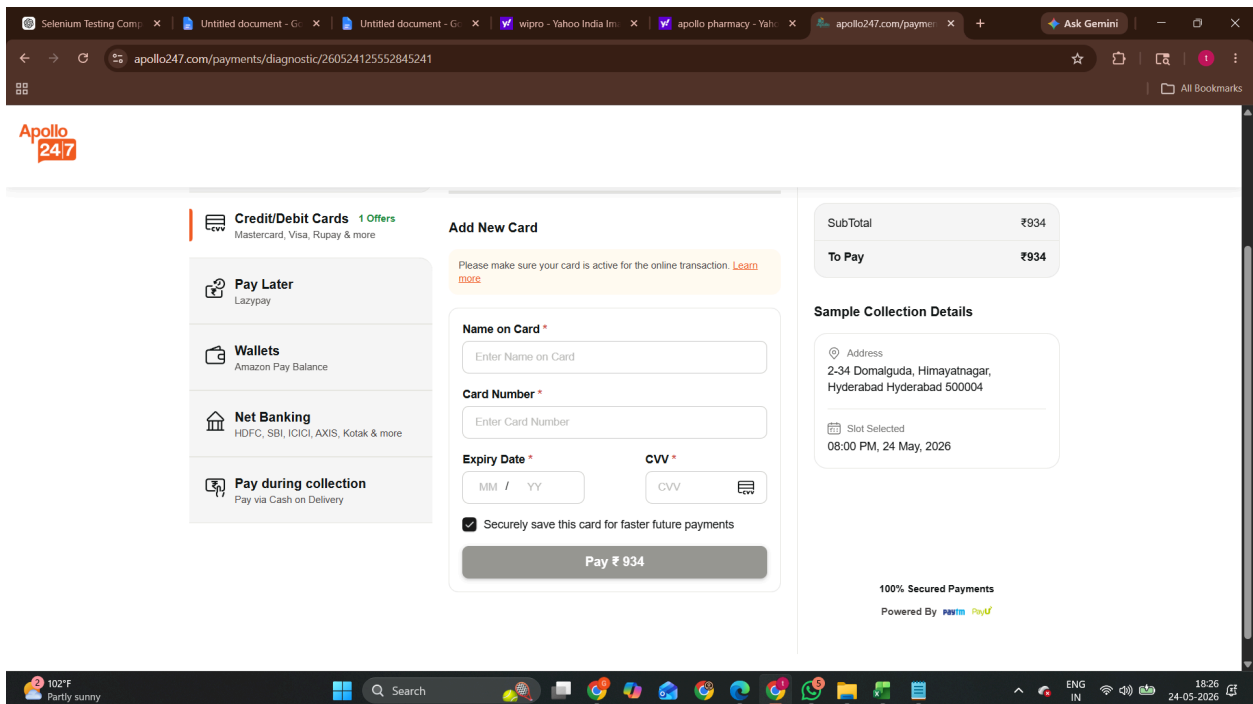


Fig10: Entering card details

Allure Reporting

Allure Reports provide professional dashboards with:

- Execution Summary
- Graphs
- Timelines
- Screenshots
- Step Execution
- Failure Analysis

Commands Used

Run Tests

```
behave -f allure_behave.formatter:AllureFormatter -o reports/allure-results
```

Generate Report

```
allure generate --output ./reports/allure-report --clean ./reports/allure-results --open
```

Conclusion

The Apollo Pharmacy Lab Test Automation Framework successfully automates a real-world healthcare web application using Selenium, Behave BDD, and Allure Reporting.

The framework demonstrates:

- Enterprise automation architecture
- End-to-End testing
- Data-driven testing
- Reporting integration
- Maintainable code structure

This project provides practical exposure to:

- Web automation
- Framework development
- Reporting tools
- BDD implementation
- Real-world testing workflows

The framework can be further enhanced using:

- CI/CD integration
- Parallel execution
- Docker support
- Jenkins pipelines
- Cloud execution
- API automation integration